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The four color theorem

Janssen, Guido

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Eindhoven University of Technology

Bachelor Final Project

The Four Color Theorem

Author:
Guido Janssen
g.janssen@student.tue.nl
0898236

Supervisor:
Judith Keijsper

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1 Introduction

Frances Guthrie was a South African mathematician who lived in the UK. He studied mathematics and Botanic studies at the University College in London. In 1852 he was trying to color the map of the UK such that two bordering counties did not have the same color. He managed to do this with four colors. Also he saw that three colors were not enough. This raised the question whether all maps could be colored with four colors. He told his brother Frederick, who was also studying mathemathics in London, about the problem. Together they presented the problem to Augustus De Morgan, who was the supervisor of Frederick and also former supervisor of Frances. In 1854 the problem was published by F.G. in the "Athenaeum". F.G. is most likely Frances or Frederick Guthrie, but this is not absolutely sure. In 1860, De Morgan published the problem again.

Alfred Kempe tried to prove the theorem in 1879 by using an unavoidable set of configurations for triangulations and his own invented Kempe-Chains. He translated all maps to planar graphs and made an algorithm that could color every minimal counterexample to the theorem with four colors, meaning that the minimal counter example does not exist. However, eleven years later, Percy Heawood found out that the algorithm of Kempe does not always work. But the work of Kempe was not useless. Heawood changed a couple of lines in the proof such that the Five Color Theorem was proven. Also the methods that Kempe used in his proof were used afterwards in many graph problems.

In 1912, the American mathematician George Birkhoff proved that every minimal counterexample to the Four Color Theorem had to be an internally 6-connected triangulation. This discovery was of great importance for the actual proof that was given in 1976 by Kenneth Appel and Wolfgang Haken [10], and later in an easier way by Neil Robertson, Daniel Sanders, Paul Seymour and Robin Thomas. Both of these proofs used the methods of reducibility, unavailability and discharging. The latter method was developed by Heinrich Heesch, who also contributed to the 1976 proof.

This report will look into the methods that were used in all attempted proofs of the Four Color Theorem. First the proof of Kempe will be explained and also it will be shown where it goes wrong. After this the methods used in the correct proof will be examined. Since the proof of 1997 uses nearly the same methods as the 1976 proof, but is much better documented and easier to understand, the 1997 proof of Robertson, Thomas, Sanders and Seymour will mostly be used to show how the mathematical tools work.

2 Kempe's proof of the Four Color Theorem

2.1 Restating the theorem

Before we can start with the proof we have to define a couple of basic concepts in graph theory.

Definition 1. *A **graph** is a pair (V, E) of sets where V is a set of vertices and E is a set of pairs of vertices of V.*

Definition 2. *A **planar graph** is a graph that can be drawn in the plane in such a way that edges do not intersect with each other except in the end points.*

Definition 3. *An **embedding** of a graph G on a surface Σ is a topological representation of a graph with each vertex represented as a point, each edge represented as a curve having the endpoints of the edge as endpoints of the curve, and no other intersections between vertices or edges.*

Definition 4. *A **plane graph** is a graph embedded in the plane.*

Definition 5. *A **face** of a plane graph G is a connected component of the subset of the plane of the embedding that is disjoint from the graph.*

Theorem 1. *In every map the countries can be colored with four colors such that two countries that border each other do not have the same color.*

Kempe used graph theory in his attempt to prove the Four Color Theorem. He associated all maps with a graph in the following way:

The vertex set V consists of all countries on the map. Then, if two countries are adjacent to each other the vertices corresponding to these countries are connected by an edge. There is one assumption that has to be made, namely that a country is a connected region. To give an example of what is meant with this assumption we could look at the USA. Alaska is part of the USA, but it is disconnected from the rest of the USA and would therefore be considered a country on its own for making the graph. It is clear that when a map is translated to a graph in this way the resulting graph is planar. Now theorem is equivalent to the following theorem:

Theorem 2. *Every planar graph has a proper 4-vertex coloring.*

Definition 6. *A vertex coloring of a graph G is called **proper** if no two vertices that are adjacent to each other have the same color.*

From now on, if we speak about a coloring we mean a proper vertex coloring.

2.2 Kempe's proof

Definition 7. *A **maximal planar graph** is a graph that is planar, but adding any edge would result in the graph not being planar anymore.*

Kempe [4, 7] realised that if he could prove that every maximal planar graph has a proper 4-coloring then he has proven that all planar graphs have a proper 4-coloring.

Kempe made use of the fact that any embedding of a maximal planar graph is a triangulation.

Definition 8. *A plane graph is a **triangulation** if all faces of the graph are triangles.*

Theorem 3. *An embedding of a maximal planar graph with more than two vertices is a triangulation.*

Proof. Assume G is a maximal planar graph that is embedded in the plane, and G is not a triangulation. Then G must have a face that has at least four vertices v_1, v_2, v_3 and v_4 in this order. But then we can connect either v_1 and v_3 or v_2 and v_4 with an edge that is embedded in only one face, meaning it does not intersect with another edge. This means that G was not maximal and so it is a contradiction. Therefore every maximal planar graph is a triangulation. □

Definition 9. *A **minimal counterexample** to the Four Color Theorem is a plane graph G with the smallest number of vertices that is not 4-colorable.*

Kempe's strategy was to take a minimal counterexample to the Four Color Theorem and then prove it was in fact 4-colorable. Therefore the minimal counterexample would not exist, and hence, no counterexample exists.

In fact, it turns out to be easy to prove that indeed a minimal counterexample to the Four Color Theorem is a triangulation.

Lemma 1. *Every minimal counterexample G to the Four Color Theorem is a triangulation.*

Proof. Let G be a minimal counterexample to the Four Color Theorem. Suppose G is not a triangulation, then G has a face of 4 or more edges. Then there are two vertices of that face that are not adjacent to each other. These can be identified. This means we remove those vertices and all edges connected to them, and then add a new vertex and connect it to all vertices that were connected to either one of the vertices. The resulting graph will be planar and loopless. Since this graph has fewer vertices than G, it can be colored with four colors. This means G can also be colored with four colors, which is a contradiction. □

After realising this, Kempe made the following proof:

Theorem 4. *Every planar graph has a proper vertex 4-coloring.*

Kempe found a set of four configurations of which at least one appears in every triangulation. He could create this set because he proved that every planar graph has a vertex of degree at most 5.

For this proof Kempe used Euler's theorem.

Theorem 5. *Let $|V|$ be the number of vertices of G, $|f|$ be the number of faces of G and $|E|$ be the number of edges of G. Then:*

$$|V| + |f| = |E| + 2, \tag{1}$$

Kempe started with proving that the following statement holds.

$$|E| \leq 3|V| - 6. \tag{2}$$

If G is a maximal planar graph embedded in the plane, all faces are triangles and all edges separate two faces. Take the sum over all faces of the edges that form the boundary of this face. Since all faces are triangles this is equal to $3|f|$. Since every edge is the boundary of two faces we counted every edge twice. This results in:

$$3|f| = 2|E|. \tag{3}$$

Substituting this in Euler's formula gives:

$$|V| + \frac{2|E|}{3} = |E| + 2. \tag{4}$$

Then,

$$3|V| + 2|E| = 3|E| + 6. \tag{5}$$

This means,

$$|E| = 3|V| - 6. \tag{6}$$

When G is not maximal the number of edges may be smaller, so the inequality holds.

Lemma 2. *Every planar graph G has a vertex of degree at most 5.*

Proof. The sum of all degrees of all vertices of G is equal to two times the number of edges. Assume all vertices have degree 6 or more, then:

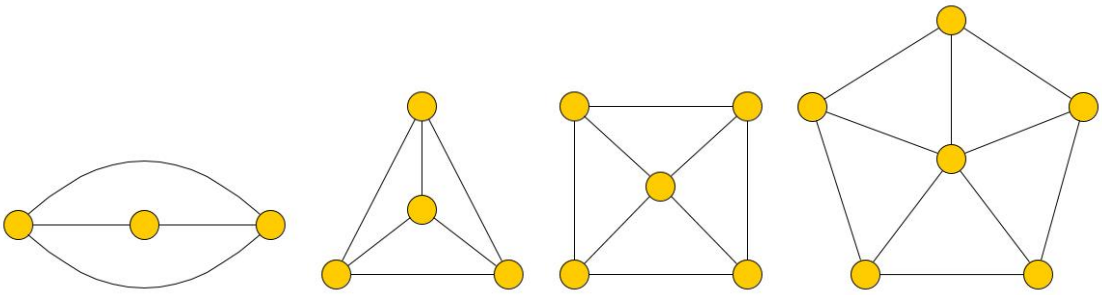
$$2|E| \geq 6|V|. \tag{7}$$

Then, from (2) and (7) it follows that:

$$2|E| \leq 6|V| - 12 < 6|V|. \tag{8}$$

This is contradicts (7), so every planar graph has a vertex of degree at most five. □

In the pictures below, it is shown what is the only way a vertex of degree 2, 3, 4 and 5 respectively can appear in a triangulation.



This means that in every triangulation with more than three vertices one of these configurations will appear. We do not have to consider graphs with three or fewer vertices since it is trivial to see that they are 4-colorable, even 3-colorable.

Now assume there is at least one planar graph that cannot be colored with four colors. Then clearly one of these graphs has to be the smallest. Remember, smallest in this case means it has a minimum number of vertices. Take this minimal counterexample and call it G.

Since G is a triangulation (proven by Lemma 1) one of the configurations above appears in G. Now remove a vertex v of degree at most 5 from G and obtain G'. Because G was a minimal counterexample G' can be colored with four colors.

The next step is to color G' with four colors and reinsert v. If v had degree 2 or 3 then there is a color available for v, since there are only 2 or 3 colors used on the neighbours of v. If v has degree 4 or 5 it becomes more difficult.

Definition 10. *A coloring Z of a circuit C in G is said to **extend** to a coloring of G if there is a coloring of G such that C is colored the same as in Z.*

Assume v has degree 4. If not all four colors have been used for the vertices around v than the coloring of G' could easily be extended to a coloring of G. If all four colors are used on the neighbours of v, then we would like to recolor one of them. This is what Kempe invented the Kempe-Chains for.

Definition 11. *Let $G = (V, E)$ be a plane graph that is k-colored with a coloring C, where $C : V \to \{1, 2, ..., k\}$. Let $i, j \in \{1, 2, ..., k\}$ and let $r \in V$, then the (i, j)-**Kempe-Chain containing** r is the maximal connected subgraph of V that contains r and in which all vertices are colored with either i or j.*

Definition 12. *A graph $G = (V, E)$ is called **bipartite** if $V = V_1 \cup V_2$ such that V_1 and V_2 are distinct, and for any two vertices that are adjacent to each other one of them is in V_1 and the other in V_2.*

Lemma 3. *A graph G is bipartite if and only if it is 2-colorable.*

Proof. Let $G = (V, E)$ be a bipartite graph, then $V = V_1 \cup V_2$. Color all vertices of V_1 with color 1, and color all vertices of V_2 with color 2. Then all vertices are colored, and by definition of a bipartite graph a vertex that is colored with color 1 is not adjacent to a vertex that is colored with color 2. Hence G is 2-colorable.

Now let $G = (V, E)$ be 2-colorable. Then the vertices of V are colored with color 1, such that a vertex that has color 1 is not adjacent to a vertex that has color 2. Now let V_1 be the set of vertices that is colored with color 1. and let V_2 be the set of vertices that is colored with color 2. Then $V = V_1 \cup V_2$ and for any two adjacent vertices one of them is in V_1 and the other in V_2. Hence G is bipartitie. □

So the (i, j)-Kempe-Chain containing r is a bipartite subgraph where V_1 consists of only vertices colored with color i and V_2 consists of only vertices colored with color j.

Lemma 4. *Let G be a graph that is k-colored. Let $i, j \in \{1, ..., k\}$. In any (i, j)-Kempe-Chain of G we can switch the colors i and j without the coloring becoming inproper, meaning there are two vertices that are adjacent to each other that have the same color.*

Proof. Take any (i, j)-Kempe-Chain in G and switch the colors i and j. Assume the coloring after switching is inproper. Then there are two vertices that are adjacent to each other that have both color i or both color j. But then they are both part of the same (i, j)-Kempe-Chain. This means they both switched colors. This means that before switching they also had the same color, which is a contradiction with the fact that the initial coloring on G was proper. □

Contraction is an act where you remove an edge, then merge the two vertices v and w that were connected to that edge, and then connect the new vertex to all vertices that were connected to v or w. It is generally known that all complete graphs with 5 or more vertices are non-planar.

Now we continue with the proof of Kempe. Assume the four neighbours of v are named a, b, c and d and are colored with colors $1, 2, 3$ and 4 respectively.

Now look at the $(1, 3)$-Kempe-Chain that contains a. There are two options. Either this Kempe-Chain contains c or it does not. If it does not, then the colors 1 and 3 can be switched within the $(1, 3)$-Kempe-Chain that contains a. This results in vertex a getting color 3 and color 1 becoming available for v.

If the $(1, 3)$-Kempe-Chain that contains a also contains c then look at the $(2, 4)$-Kempe-Chain that contains b. If this Kempe-Chain does also contain d it means there is a path from a to c with only vertices that are colored with color 1 and 3, and also a path from b to d with only vertices that are colored with color 2 and 4. In any embedding of G these two paths will intersect with each other. Since we assumed G was planar this intersection has to take place in a vertex. Let us name this vertex w. Now w is a part of the $(1, 3)$-Kempe-Chain that contains a and c, but also of the $(2, 4)$-Kempe-Chain that contains b and d. But then w needs to be color 1 or 3, and at the same time color 2 or color 4. This cannot happen since every vertex can have only one color. Therefore it gives a contradiction, so

the $(2,4)$-Kempe-Chain that contains b does not contain d. Then we can switch color 2 and 4 in this Kempe-Chain, which results in b getting color 4. Then color 2 becomes available for v.

So in both cases a 4-coloring of G' can be extended to a 4-coloring of G.

Now we have proven the following lemma.

Lemma 5. *A minimal counterexample to the four color theorem cannot have a vertex of degree 4 or less.*

Next, we will look at the case in which v has degree 5 and assume all four colors are used on for the vertices a, b, c, d and e around v. This can be done in only one way, namely 1, 2, 3, 4, 2. All other ways to color these five vertices with four colors are essentially the same. This is true since colors can always be permuted without the coloring becoming inproper. Again remove v to obtain G'. Color G' with four colors and reinsert v. Take the $(1,3)$-Kempe-Chain that contains a. If this does not contain c the colors 1 and 3 can be switched in the chain and color 1 becomes available for v.

If the $(1,3)$-Kempe-Chain that contains a does contain c then look at the $(1,4)$-Kempe-Chain that contains a. If this does not contain d then the colors 1 and 4 can be switched and color 1 becomes available for v.

So far everything went fine, but the next argument Kempe used in his attempted proof will be wrong in general. He argued as follows.

If both the $(1,3)$-Kempe-Chain that contains a contains c and the $(1,4)$-Kempe-Chain that contains a contains d then look at the $(2,4)$-Kempe-Chain that contains b. This cannot contain d, because then again the paths will intersect each other in a vertex that then would need to have two colors at the same time. Similarly the $(2,3)$-Kempe-Chain that contains e cannot contain b and c. Therefore we can simultaneously switch the color 2 and 4 in the $(2,4)$-Kempe-Chain that contains b and the colors 2 and 3 in the $(2,3)$-Kempe-Chain that contains e. Now color 2 has become available for v. Hence G is 4-colorable.

This way Kempe concluded the minimal counterexample can be colored with four colors, and thus all planar graphs can be colored with four colors.

2.3 The mistake

Kempe's proof was assumed to be correct until in 1890 Percy Heawood found a counterexample [9]. He found a graph of 25 vertices with a coloring of the graph and a vertex of degree 5, such that if you would use Kempe's algorithm would have two same colored vertices connected to each other. In the picture below you can see the graph Heawood used. The black vertex is the vertex v of degree 5 that was removed from the graph to color the resulting graph. The red vertex will be considered a, and then we will rotate clockwise to assign b, c, d and e. The second graph shows the result of applying Kempe's algorithm to the graph. Here blue and green are switched in the (green,blue)-Kempe-Chain that contains b, and yellow and blue are switched in the (yellow,blue)-Kempe-Chain that contains e. In the second graph the top vertex and the lower right vertex are connected and are both blue, meaning this is not a properly colored graph, and Kempe's argument is not valid for all cases.

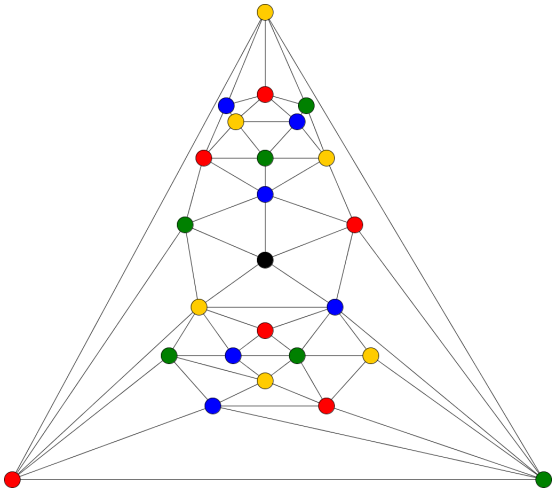


Figure 1: Heawood's counterexample before applying Kempe's algorithm

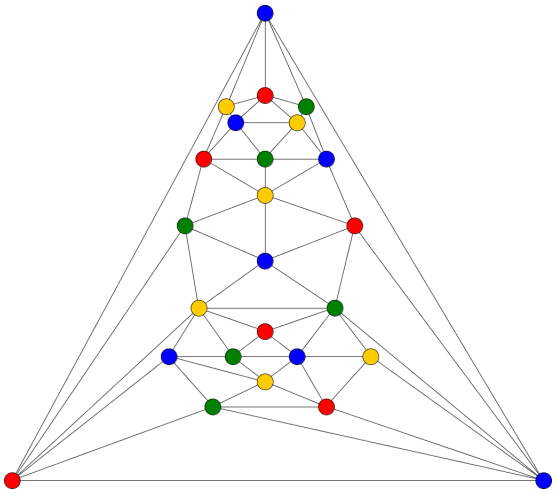


Figure 2: Heawood's counterexample after applying Kempe's algorithm

After Heawood found this counterexample to Kempe's proof, other authors discovered other much smaller graphs that also could be used as a counterexample to Kempe's proof. One of them is in the figure below. Again, the red vertex will be considered a, and then we will rotate clockwise to assign b, c, d and e. Again blue and green are switched in the (blue,green)-Kempe-Chain that contains b, and blue and yellow are switched in the (blue,yellow)-Kempe-Chain that contains e.

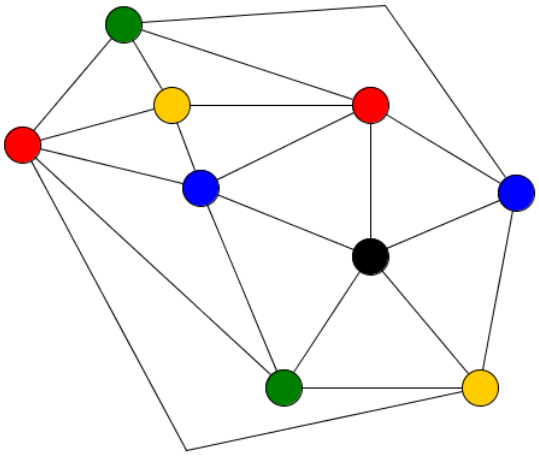


Figure 3: Smaller counterexample before applying Kempe's algorithm

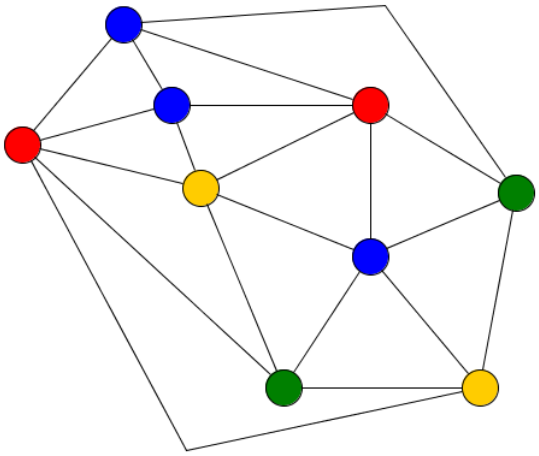


Figure 4: Smaller counterexample after Kempe's algorithm

Although Heawood had found a flaw in the last part of the proof of Kempe it was not useless. Heawood managed to prove the Five Color Theorem by changing some small things to the last case of Kempe's proof. Also some of the techniques used in Kempe's proof became very handy tools in graph theory.

2.4 The Five Color Theorem

Theorem 6. *Every planar graph has a proper 5-coloring.*

The proof of the Five Color Theorem starts exactly the same as the proof Kempe gave for the Four Color Theorem. The only difference is the proof of the case when vertex v has degree 5. This is were it went wrong with Kempe's proof, because swapping two Kempe-Chains simultaneously could result in a non-proper coloring. If v has degree 2, 3 or 4, Lemma 5 already shows that four colors are sufficient.

If v has degree 5 and the five neighbours of v are colored with all five available colors. Then a same argument as with the degree 4 case of Kempe's proof works. Look at the $(1,3)$-Kempe-Chain that contains a. If this does not contain c the colors 1 and 3 can be swapped in the $(1,3)$-Kempe-Chain that contains a. If it does contain both a and c look at the $(2,4)$-Kempe-Chain that contains b. Because of planarity this does not contain d so now the colors 2 and 4 can be swapped in the $(2,4)$-Kempe-Chain that contains b. So in both situations one color becomes available for v, so the graph has a proper 5-coloring. Hence, all planar graphs have a proper 5-coloring.□

3 Birkhoff's proof concerning the properties of minimal counterexamples

In 1913, the American mathematician George Birkhoff managed to prove that every minimal counterexample to the Four Color Theorem has to be an internally 6-connected triangulation [8]. The proofs in this section are based on the proofs Steinberger gave in his article [1].

Definition 13. *In a graph G, for any set $K \subset V$, $G\backslash K$ is the graph that is made by deleting all vertices in K from G. So the vertex set V consists of all vertices of G that are not in K, and the edge set E consists of all edges between two vertices that are both not in K.*

Definition 14. *A graph $G = (V, E)$ is **k-connected** if deleting any set $K \subset V$ of size smaller than k would result in the graph $G\backslash K$ being connected.*

Definition 15. *A graph G is **internally k-connected** if for any set $K \subset V$ with $|K| \leq k-2$ the graph $G\backslash K$ is connected, and for $|K| = k-1$ the graph $G\backslash K$ is connected, or consists of two components of which one consists of only 1 vertex.*

Definition 16. *In a plane graph G the **interior/exterior** of a circuit is the open region inside/outside the closed curve that represents the circuit in the embedding of the graph.*

Definition 17. *A **short circuit** of a plane graph G is a circuit C in G of length 5 or smaller such that both the interior and the exterior of C contain at least one vertex of G if $|C| \leq 4$ or at least two if $|C| = 5$.*

Definition 18. *Let G be a graph and $K \subset V(G)$ then $G(K)$ is the subgraph induced by $K \subset V$, such that $G(K) = (K, \{\{k_1, k_2\} \in E(G) | k_1, k_2 \in K\})$.*

Lemma 6. *A plane triangulation G is internally 6-connected if and only if G does not have a short circuit and every vertex has degree at least 5.*

Proof. Part 1:

Let G be a plane internally 6-connected triangulation. Then every vertex has degree at least 5, because otherwise one could delete the neighbours of a vertex of degree smaller than 5. The resulting graph would be disconnected while there were fewer than 5 vertices deleted, which gives a contradiction with G being internally 6-connected. Assume G has a short circuit C of length 5 or less. Now if we create G' by deleting C then G' is disconnected, since both the interior and exterior of C are non-empty. If C had length at most 4 this is a contradiction to G being internally 6-connected. So C has length 5. By definition a short circuit of length 5 has at least two vertices in both the interior and the exterior of C. This means that G is not internally 6-connected, so this is a contradiction. So if G is an internally 6-connected triangulation then G does not have a short circuit and G and has minimum degree 5.

Part 2: We will give a sketch of the proof. Let G be a plane triangulation in which all vertices have degree at least 5, and in which no short circuit appears. Assume G is not internally 6-connected. This means we can either find a set K such that $G\backslash K$ is disconnected when $|K| \leq 4$, or $G\backslash K$ consists of two components of both at least two vertices when $|K| = 5$.

An embedded triangulation always has three vertices, say v_1, v_2 and v_3, that are incident with the infinite region. They form a triangle face in which all other vertices are embedded.

Because G is a triangulation, $G(K)$ either has to be a circuit that separates the interior and the exterior of the circuit, or $G(K)$ is a path between two vertices on the infinite face.

Let $G(K)$ be a circuit that separates the interior an the exterior. Then if $|K| \leq 4$ both the interior and exterior of $G(K)$ contain at least one vertex. But then $G(K)$ is a short circuit, which is a contradiction with the fact that G does not contain a short circuit. If $|K| = 5$ then both the interior and the exterior

of $G(K)$ need to contain at least two vertices, otherwise G would be internally 6-connected. But then again $G(K)$ is a short circuit.

Let $G(K)$ be a path from v_1 to v_2 where v_1 and v_2 are on the infinite face. since the infinite face has only three vertices the only way to make a separation is to add vertices from G to $G(K)$. But then we would create a circuit. Then again if $|K| \leq 4$ both the interior and exterior of $G(K)$ contain at least one vertex. But then $G(K)$ is a short circuit, which is a contradiction with the fact that G does not contain a short circuit. If $|K| = 5$ then both the interior and the exterior of $G(K)$ need to contain at least two vertices. But then again $G(K)$ is a short circuit.

Therefore, if no short circuit appears in a triangulation G and every vertex of G has degree at least 5, then G is internally 6-connected. □

Theorem 7. *Every minimal counterexample to the Four Color Theorem is an internally 6-connected triangulation.*

Proof. Let G be an embedding of a minimal (nonempty) counterexample to the Four Color Theorem. We already showed with Lemma 1 that this has to be a triangulation. Furthermore Kempe already showed that every vertex in a minimal counterexample to the Four Color Theorem has at least degree 5. This was shown in Lemma 5.

Next, we will show that G does not have a short circuit. First suppose G has a short circuit C of length 3.

Because G is minimal, G minus the interior of the short circuit has a proper 4-coloring, as has G minus the exterior of the short circuit. Now both the interior and the exterior have a coloring on C. Since C has length 3 it has to be colored with three different colors. Then we fix a coloring on C that extends to a coloring of G minus the interior of C. Without loss of generality this assigns $C = (a, b, c)$ with colors 1,2 and 3 in that order. Now look at a coloring of G minus the exterior of C. This has also a coloring on C. If this is not (1,2,3) then we can just permute colors such that (a,b,c) is colored with (1,2,3). Now we found a coloring of C that extends to both the interior and the exterior of C, meaning G is 4-colorable.

Now look at the case that G has a short circuit C of length 4 with vertices (a, b, c, d). Let A be the graph consisting of G minus the interior of C and B be the graph consisting of G minus the exterior of C. Since A and B are both smaller than G they can be colored with four colors $1, 2, 3$ and 4 because G is a minimal counterexample. Then $C = (a, b, c, d)$ can be colored in four different ways: (1,2,3,4), (1,2,1,3), (1,2,3,2) and (1,2,1,2) modulo permutations of the colors and cyclic permutations of (a, b, c, d).

Create A' by adding an edge between a and c in A, so there must be a coloring of A' in which a and c have different colors since A' is smaller than G. This means there also is a coloring of A in which a and c have different colors. The same holds for b and d when A'' is created by adding an edge between b and d. This means that the coloring (1,2,3,4) could be a possible coloring of C in A. If this is not the case the coloring (1,2,1,3) of C extends to A when an edge is added between b and d, and (1,2,3,2) extends to A when an edge is added between a and c. So either the coloring (1,2,3,4) of C extends to A, or both the colorings (1,2,1,3) and (1,2,3,2) extend to A.

Equivalently this can be shown for B, which means that either the coloring (1,2,3,4) extends to B or both (1,2,1,3) and (1,2,3,2) extend to B.

If the same coloring of C extends to A and B we are done and we found a proper coloring for both G minus the interior of C and G minus the exterior of C in which the coloring of C is the same. This means this coloring of C also extends to G.

If no coloring of C extends to both A and B, we have either C can be colored with (1,2,3,4) in A and with both (1,2,1,3) and (1,2,3,2) in B or the other way around. Without loss of generality we can assume that (1,2,3,4) extends to A, and both (1,2,1,3) and (1,2,3,2) extend to B. Now we can use a similar argument as Kempe did in his proof when v had degree 4. Look at the $(1, 3)$-Kempe-Chain in A that contains c. If this does not contain a switch colors 1 and 3 in the chain, otherwise look at the $(2, 4)$-Kempe-Chain

that contains b, this does contain d so switch color 2 and 4. Hence the coloring of C in A can be changed to either (1,2,1,4) or (1,2,3,2) which both extend to B, meaning G was not a counterexample. Hence no counterexample is minimal, so there is no counterexample.

Finally look at the case that G contains a short circuit C of length 5 with vertices a, b, c, d and e. Again let A be G minus the interior of C and let B be G minus the exterior of C. Now, both A and B have a proper 4-coloring because they have fewer vertices than G. Then, $C = (a, b, c, d, e)$ can be colored in only two different ways up to rotation and permutation of the colors, namely P = (1,2,1,2,3) and Q = (1,2,1,4,3). Let $P_1...P_5$ be the equivalence classes under rotation of P and $Q_1...Q_5$ be the equivalence classes of Q. The following notation will be used: $H|P_i$ means that a coloring from the equivalence class P_i on C can be extended to a coloring of H. Here H is either A or B. We will prove the following two statements:

1. $H|P_i \wedge \neg H|P_{i+1} \implies H|Q_i$
2. $\neg H|P_{i+1} \wedge \neg H|Q_{i+1} \implies H|P_i$.

To prove statement 1 we can without loss of generality assume that $i = 1$ and $H = A$. So in particular, the coloring (1,2,1,2,3) from P_1 does extend, but the coloring (3,1,2,1,2) from P_2 does not. A can be colored with four colors, since it has fewer vertices than G. Then look at the $(1, 3)$-Kempe-Chain in A that contains c. If this does not contain a then the colors 1 and 3 can be swapped in that $(1, 3)$-Kempe-Chain, but then the coloring (1,2,3,2,3) from P_2 extends to A, which is a contradiction. Note that if the $(1, 3)$-Kempe-Chain that contains c does contain e, then it also contains a, because a and e are adjacent to each other. So the color of e remains 3. Therefore the $(1, 3)$-Kempe-Chain that contains c does contain a, but then the $(2, 4)$-Kempe-Chain that contains b does not contain d so if we swap colors 2 and 4 on the $(2, 4)$-Kempe-Chain that contains d we get (1,2,1,4,3), which is in Q_1.

For statement 2 we again say without loss of generality $i = 1$ and $H = A$. Now we identify a and c, so we remove those vertices and all edges connected to them, and then add a new vertex and connect it to all vertices that were connected to either a or c (or both). If a and c were adjacent in A then identifying them would create a loop. However, this is not possible since then A would have a short circuit of length 3 or 4. The reason for this is that both the inner and outer region of C have at least two vertices of G in it. The resulting graph is has fewer vertices than A, so it has a 4-coloring. This means that there is a 4-coloring of A in which a and c have the same color. This implies that either a coloring from P_1 or P_5 or Q_1 has to extend to A, since these are the only ones were a and c have the same color. So this means $\neg A|P_1 \wedge \neg A|Q_1 \implies A|P_5$. The same can be shown for all i equivalently.

Now make a new new graph A' from A by adding a vertex in the interior of the circuit, and connect it to all vertices of C. This graph is planar, has no loops and is smaller than G, because by definition of a short circuit there are at least two vertices in the interior of C if C has length 5. This means A' is 4-colorable, so A must have a coloring of C were only three colors are used on C. Hence P_i extends to A' for some i, so it also extends to A.

Likewise we can show that also P_j extends to B for some j. Now assume none of the equivalence classes of P and Q extend to both A and B. So there must be at least one i such that P_i extends to A, but P_{i+1} does not, otherwise $A|P_i$ would be true for all i. Without loss of generality choose $i = 1$, meaning that $A|P_1$ and $\neg A|P_2$. Now the first statement implies $A|Q_1$. Statement two used on B with $i = 5$ gives us $\neg B|P_1 \wedge \neg B|Q_1 \implies B|P_5$. Use the first statement again to get $B|P_5 \wedge \neg B|P_1 \implies B|Q_5$. This means that both P_5 and Q_5 do not extend to A. Then the second statement gives $\neg A|P_5 \wedge \neg A|Q_5 \implies A|P_4$. We repeat the same argument until we reach $A|P_2$, which is a contradiction.

This means every minimal counterexample to the Four Color Theorem does not have a short circuit. □

4 Reducibility

Reducibility[2][3] is one of the most important concepts in the proofs of 1976 and 1997. The goal is to find a set of configurations of which we can show that if this configuration appears in a minimal counterexample, it can be colored with four colors. This results in the conclusion that the minimal counterexample does not exist, so the theorem is true.

To prove that a configuration is reducible it is needed to define a configuration, but first we will define a near-triangulation.

Definition 19. *A connected plane graph G is a **near-triangulation** if every face of G is a triangle, except for possibly the infinite region.*

Definition 20. *Let G be a plane graph, then a configuration consists of $K \subset V$, where $G(K)$ is near-triangulation, and a map $\gamma_K : V(G(K)) \to \mathbb{N}$ which has the following properties:*

- *For every vertex v, $G(K)\backslash v$ has at most two components, and if there are two then $\gamma_K(v) = d(v)+2$ were $d(v)$ is the degree of v in $G(K)$.*
- *For every vertex v, if v is not incident with the infinite region then $\gamma_K(v) = d(v)$. Otherwise $\gamma_K(v) > d(v)$. In either case, $\gamma_K \geq 5$.*
- *$\sum_v (\gamma_K - d(v) - 1) \geq 2$ summed over all vertices v that are incident with the infinite region such that $G(K)\backslash v$ is connected.*

Definition 21. *$\sum_v (\gamma_K - d(v) - 1)$ summed over all vertices v that are incident with the infinite region such that $G(K)\backslash v$ is connected is called the **Ring Size** of K.*

Intuitively $\gamma_K(v)$ can be seen as the degree of v if K would appear in a larger triangulation. This can be seen since $\gamma_K(v)$ is equal to $d(v)$ if v is not on the boundary of K, which means that in a larger triangulation v would have the same neighbours. If v is on the boundary of K then $\gamma_K(v) > d(v)$ and $\gamma_K(v) \geq 5$.

Definition 22. *A near-triangulation S is called a **free completion** with ring R of a configuration K if:*

- *R is an induced circuit of S and R bounds the infinite region of S.*
- *$G(K)$ is an induced subgraph of S.*
- *$V(G(K)) = S\backslash V(R)$.*
- *The infinite region of K includes $V(R)$ and $E(R)$.*
- *Every v in S that is not in $V(R)$ has degree $\gamma_K(v)$.*

Moreover, if there are two free completions S_1 and S_2 of K then there is a homeomorphism of the plane that maps every vertex of S_1 to a distinct vertex of S_2 and every edge of S_1 to a distinct edge of S_2, such that $G(K)$ is fixed pointwise. This is because the number of vertices and the number of edges of the free completion of a certain configuration K is equal for all possible free completions.[5].

Definition 23. *A configuration K is **reducible** if K cannot appear in a minimal counterexample to the Four Color theorem.*

In the 1997 proof there are two different types of reducibility that are used, namely C-reducibility and D-reducibility.

Definition 24. *Let G be a minimal counterexample to the Four Color Theorem. Let S be a free completion*

of a configuration K that appears in G with ring R. Let C' be the set of all 4-colorings of R that extend to a 4-coloring of $G\backslash V(K)$. The configuration K is said to be ***D-reducible*** *if $C' = \emptyset$.*

So if C' is empty then there is no 4-coloring of R that extends to a 4-coloring of $G\backslash V(K)$. Clearly this cannot happen because $G\backslash V(K)$ is smaller than G, which was a minimal counterexample. So a D-reducible configuration cannot appear in a minimal counterexample.

The only problem we got is that S is not necessarily a subgraph of G. However this is not a problem because some vertices of the ring can be identified in G. Now $G\backslash V(K)$ is smaller than G so since G is a minimal counterexample $G\backslash V(K)$ can be 4-colored.[1]

To show a configuration is D-reducible we assume C' is nonempty. Then we construct C, which is the set of all colorings of R that extend to a 4-coloring of $S = G(K) + V(R)$. Let C^* be the set of all 4-colorings of R. Then $C' \subset C^* - C$. Otherwise there is a 4-coloring of R that extends to both S and $G\backslash G(K)$, but then G is 4-colorable. Since we assumed G was a minimal counterexample this cannot happen, so indeed $C' \subset C^* - C$. After that you have to show that a coloring in $C^* - C$ does not belong to C'. If you have done this, you have shown C' is empty, which is a contradiction. so then you have shown the configuration is D-reducible.

Definition 25. *Let G be a minimal counterexample to the Four Color Theorem. Let S be a free completion of a configuration K that appears in G with ring R. Let C' be the set of all 4-colorings of R that extends to a 4-coloring of $G\backslash V(K)$. Then K is said to be* ***C-reducible*** *if there exists a near-triangulation S', called a reducer, obtained from S by replacing $G(K)$ by a smaller graph (and possibly identifying some vertices of R to get R') such that C' becomes empty.*

So in the case of D-reducibility $G(K)$ is deleted from G to obtain a smaller graph that is 4-colorable. However we could also replace K by some smaller graph to make sure the new graph G' is 4-colorable. Here the problem rises that we must check whether G' is loopless, which can be very difficult. This was a major issue of Appel and Haken in their proof in 1976. If the resulting graph is a triangulation the reducer is called a safe reducer. In the 1997 proof they only looked at the case were a maximum of four edges are contracted. In this case, checking safety is very easy.

In all (correct) proofs that have been published today a computer is used to check whether a configuration is reducible. This is needed because the sets of configurations that have to be checked on reducibility contain such large configurations that checking all colorings by hand would not be possible. For example, if a configuration has ring-size 14 then approximately 200.000 colorings have to be checked. All known proofs use a set of configurations of which they can show (with the help of the computer) that they are reducible. Finally, they will show that in every internally 6-connected triangulation one of these configurations will appear. The set used in the 1976 proof of Appel and Haken consisted of 1476 configurations. Robertson, Seymour, Sanders and Thomas managed in the 1997 proof to reduce this set to 633 members. In 2010 John Steinberger managed to proof the theorem with a set of 2822 configurations that were all D-reducible, but eventually this needed even more computer time. Not only because the size of the sets and the number of discharging rules, but also because of the size of the configurations he needed.

5 Unavoidability and Discharging

5.1 The method of unavoidability and discharging

Discharging is a method, developped by Heinrich Heesch, that is used to show unavoidability of a configuration. The authors of the 1997 proof wrote multiple articles [2, 5, 6] on how they used discharging to prove unavoidability. We will follow the ideas given in these articles.

Definition 26. *A set of configurations is called* ***unavoidable*** *if at least one of its elements appears in every minimal counterexample to the Four Color Theorem.*

Since we already proved that every minimal counterexample is an internally 6-connected triangulation this means a set of configurations is certainly unavoidable if one of its elements appears in every internally 6-connected triangulation.

To understan the discharging method we need to show that the following is true:

$\sum_{v \in G} (6 - d(v)) = 12$.

Proof. We will use three formulas:

1. $\sum_{v \in G} d(v) = 2|E|$,
2. $2|E| = 3|F|$,
3. $|V| + |F| = |E| + 2$ (Euler's formula).

The first statement is true because every edge is incident with exactly two vertices. The second statement is true because every face of G is a triangle and every edge is incident with two faces.

Now, $\sum_{v \in G} (6 - d(v)) = \sum_{v \in G} 6 - \sum_{v \in G} d(v) = 6|V| - 2|E| = 6|V| - 3|F| = 6(|E| - |F| + 2) - 3|F| = 6|E| - 9|F| + 12 = 6|E| - 6|E| + 12 = 12$.

This means that every triangulation G in which every vertex has degree 5 or more, there must be at least 12 vertices of degree 5. When G has vertices of degree 7 or more then this number will be higher. □

Definition 27. *A* ***good configuration*** *K is a configuration that is an element of an unavoidable set.*

Definition 28. *The* ***second neighbourhoud*** *of a vertex v consists of all vertices w such that there is a path from v to w of length at most 2.*

First we will give a charge of $10(6 - d(v))$ to every vertex. From the formula above we know that the total charge of the graph will be 120. Then a list of discharging rules is used to redistribute the charges over the vertices. In the 1976 proof 486 discharging rules were used, whereas in the 1997 proof "only" 32. Therefore we will focus on the 1997 proof. After the rules have been applied all vertices have a new charge, but the total amount has not changed. Since the total charge is positive, there must be at least one vertex that has a positive charge. At the end we will show that a good configuration appears in the second neighbourhood of this vertex.

First we will give formal definitions for a rule and a pass. After that we will look at the case that a vertex v that ends up with positive charge has either degree 6 or smaller, or degree 12 or higher. These two cases are not too difficult to prove by hand. The cases that v has degree 7,8,9,10 or 11 require too much work to do by hand and therefore they need to be handled separately by a computer to give a proof that would in total be approximately 13.000 lines.

Definition 29. *A* ***rule*** *is a sixtuple $(G, \beta, \delta, r, s, t)$ where:*

- *G is a near-triangulation*

- *β is a map from $V(G)$ to $\mathbb{N}$*
- *δ is a map from $V(G)$ to $\mathbb{N} \cup \{\infty\}$*
- *r is an integer*
- *s and t are distinct vertices of G that are adjacent to each other*

So intuitively a rule can be seen as a graph in which two numbers are assigned to each vertex and one or two arrows are added to a certain edge in the graph, meant to send charge from s to t.

Definition 30. *A **Pass** is a quadruple (K, r, s, t) such that:*

- *K is a configuration*
- *r is a positive integer*
- *$s, t \in G(K)$ are distinct*
- *For each $v \in P$ there is a v, s-path and a v, t-path of length 1 or 2.*

Definition 31. *A pass P is said to **Obey** a rule $(G, \beta, \delta, r, s, t)$ if P is isomorphic to some (K, r, s, t) where $G = G(K)$ and $\beta(v) \leq \gamma_(v) \leq \delta(v)$ for all v in G.*

In this the last definition γ_K is mentioned. As K is a configuration every vertex has its own $\gamma_K(v)$, which represents the degree that v would have if K would appear in a triangulation. For the rest of this chapter if we speak about the degree of a vertex v we mean $\gamma_K(v)$.

To get a better idea of what a rule is we can draw them. The following conventions are used when drawing a vertex of a configuration.

●	$\gamma_K(v) = 5$
·	$\gamma_K(v) = 6$
○	$\gamma_K(v) = 7$
□	$\gamma_K(v) = 8$
▽	$\gamma_K(v) = 9$
⬠	$\gamma_K(v) = 10$

There are three important cases that can appear with the β and δ maps, namely:

1. $5 \leq \beta(v) = \delta(v) \leq 8$
2. $\beta(v) = 5, 6 \leq \delta(v) \leq 8$
3. $5 \leq \beta(v) \leq 8, \delta(v) = \infty$

In a drawing case 2 will be represented with a minus sign added to v, case 3 will be represented with a plus sign added to v. For case 1 there is nothing added to the vertex. If there is no sign added to v then $\beta(v) = \gamma_K(v) = \delta(v)$, meaning there is only one possibility for the degree of v.

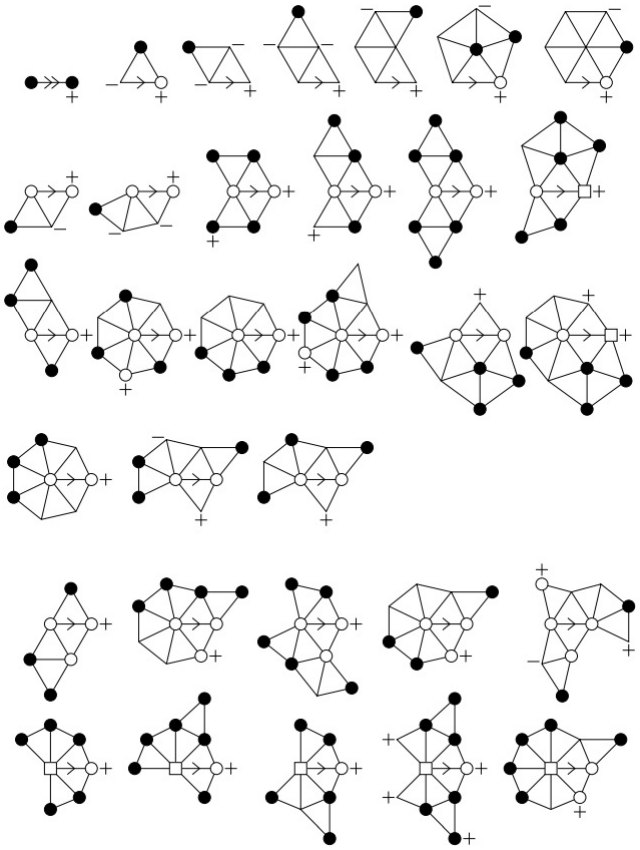
If there is a minus sign attached to a vertex the dot represents the value of $\delta(v)$, meaning this is the upper bound for the degree of v (the lower bound is 5). If there is a plus sign attached to a vertex then the dot represents the value of $\beta(v)$, meaning it is the lower bound for the degree of v.

The cases that appear in the rules of the 1997 proof are listed in the table below.

Symbol	$\beta(v)$	$\delta(v)$	Possible degree of v
$\bullet$	5	5	5
$\bullet+$	5	∞	≥ 5
$\cdot$ -	5	6	5 or 6
$\cdot$	6	6	6
$\cdot+$	6	∞	≥ 6
$\circ$	7	7	7
$\circ+$	7	∞	≥ 7
$\square$	8	8	8
$\square+$	8	∞	≥ 8

Table 1: Conventions

The figure below shows the drawings of all 32 rules that were used in the 1997 proof. The rules are in ascending order of the degree of the source. If this is equal they are listed in ascending order of the sink. All rules that have equal degree of the source and sink are listed in random order.



Every pass obeys exactly one rule. We call $r(P)$ the value of the pass, s is the source and t is the sink. We say a pass P appears in G if $K(P)$ appears in G. Then the charge of a vertex v will be:

$10(6 - d(v) - (\sum r(P) : P$ appears in G, v is the source of $P) + (\sum r(P) : P$ appears in G, v is the sink of $P)$, where the sum is taken over all passes P that appear with v as either source or sink.

This can be seen as the original charge plus all charge that a vertex v gets from being sink in passes, minus all charge it loses from being source in passes.

5.2 Case 1: v has degree 5 or 6.

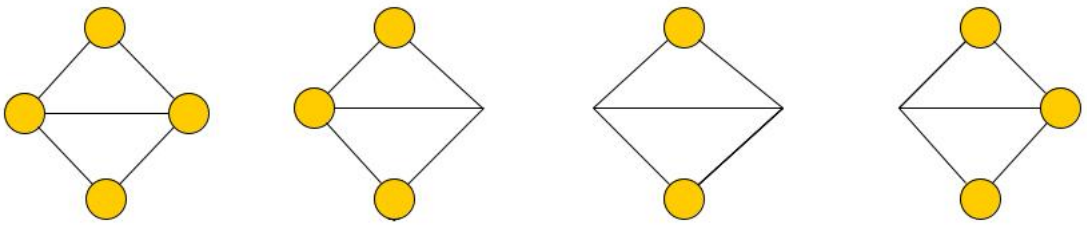
To prove that a good configuration must appear in the second neighbourhood of a vertex that has positive charge we will split up in multiple cases. First we will look at the case that the vertex v that ends up with positive charge is of degree 5 or 6. In the proof we will make use of the fact if two vertices v and w are adjacent to each other in a triangulation, then there are exactly two vertices that are adjacent to both v and w. We will first proof the following lemma:

Lemma 7. *Let v be a vertex that ends up with charge $N > 0$ and has degree 5 or 6. Let p_i be the sum of $r(P)$ over all passes P that obey rule i and have sink v, and let q_i be the sum of $r(P)$ over all passes that obey rule i and have source v. Furthermore, let no good configuration appear in the second neighbourhood of v. Then the following equations hold:*

1. $p_1 = q_2 + q_3$
2. $p_3 = q_4$
3. $p_4 = q_5 + q_6$
4. $p_5 = q_7$.

Proof. First, let v have degree 5 and let (x, y, z) be a triple of distinct vertices that are adjacent to v, such that both x and z are adjacent to y and the degree of x is 5. Let X be the set of all these triples. The idea of the proof is that we count the number of triples in two different ways. We know v has degree 5 or 6. In both cases rule 1 is applicable, since it sends two charge from every vertex of degree 5 to a vertex of degree at least 5 (because of the plus sign). So the charge that arrives in v is equal to two times the number of vertices of degree 5 that are connected to v. This is equal to $|X|$, since by the construction of X every vertex of degree 5 is in two triples.

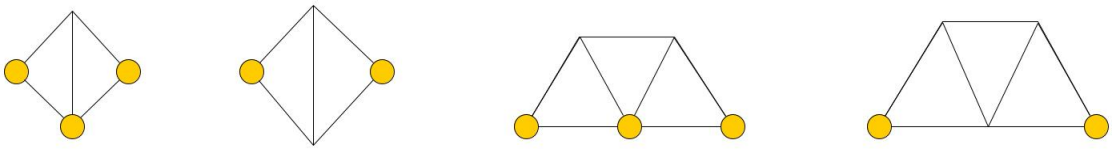
Secondly, we will look at q_2 and q_3. Rule 2 sends charge from a vertex of degree 5 or 6 to a vertex of degree at least 7. Furthermore, both vertices are connected to a vertex of degree 5. Since we are looking at q_2 vertex v is the source. Say q_2 is the number of triples $|X_1|$. Then X_1 is the set of (x, y, z) where x has degree 5 and y has degree 7 or higher. Now look at rule 3. The same way as done with q_2 we can see that $q_3 = |X_2|$ is equal to the number of triples were x has degree 5, y has degree 6 or less, and z has degree 6 or higher. Finally, $|X_3|$ is the set of triples where the degree of x is 5, the degree of y is 6 or less and the degree of z is 5. Now by construction $|X| = |X_1| + |X_2| + |X_3|$. In the picture below you can see how the triples of X_3 would look. In all cases v is the most left vertex.



If we take a close look at the list of good configurations we see all four of these configurations appear (conf(1, 1, 1), conf(1, 1, 2), and conf(1, 1, 4), where conf(x,y,z) refers to page x, row y, column z of the list of configurations in the paper of Robertson, Sanders, Seymour and Thomas [6])) and so, since we assumed that no good configuration appeared, this cannot happen. But this means that $|X_1| + |X_2| = q_2 + q_3$ is equal to the number of triples which is $|X|$. So indeed $p_1 = q_2 + q_3$.

If we look at rule 3,4 and 5, we see that charge is sent to a vertex of degree 6 or higher. Therefore, if the degree of v is 5 we know that p_3, p_4 and p_5 are all zero. Furthermore in rule 5,6 and 7 we also see that the degree of the source has to be at least 6, meaning that q_5, q_6 and q_7 are all 0. This means all statements would hold. So we will now assume the degree of v is 6. First we will define a different kind of triple.

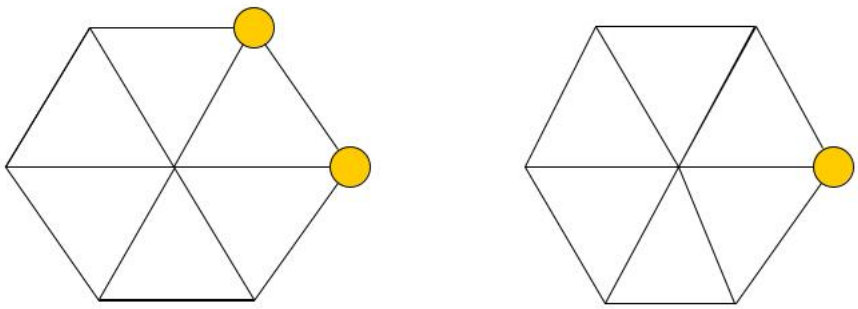
From now on (x, y, z) is called a triple if x, y and z are all distinct neighbours of v, y is adjacent to both x and z, the degree of x is 5 or 6, the degree of y is 5 or 6 and the degree of u is 5. Here u is the vertex different from v that is adjacent to both x and y. The set of all triples is called Y. With this definition p_3 is equal to $|Y|$, because v is the source and has degree 6, x and y have both degree 5 or 6 (because of the minus sign) and the degree of the vertex adjacent to x and y is indeed 5. So, to prove the statement, q_4 also has to be equal to $|Y|$. Now say $|Y| = |Y_1| + |Y_2|$. We define Y_1 as all triples (x, y, z) where the degree of x is 5 or 6, the degree of y is 5 or 6, the degree of z is 6 or higher, and the degree of u is 5. This implies that $Y_2 = Y - Y_1$. This is the set of all triples, except for those where the degree of z is equal to 5. Look at the triples that would be in Y_2. If Y_2 is non-empty one of the following configurations would show up:



Again, if we take a look at the list of good configurations we can see that all these are included, namely conf(1, 1, 2), conf(1, 1, 4), conf(1, 1, 5) and conf(1, 1, 7). Therefore, since we assumed no good configuration could appear, the degree of z has to be at least 6 and $q_4 = |Y| = p_3$, which proves the statement.

Statement 3 says $p_4 = q_5 + q_6$. Now define a triple as follows: (x, y, z) is a triple if x, y and z are adjacent to v, y is adjacent to both x and z, x has degree 6, y has degree 5 or 6, u has degree 5 or 6 and w has degree 5. Here u is the vertex adjacent to x and y that is not v, and w is the vertex adjacent to y and u that is not x. The set of all triples will be called Z. Now we can see that $p_4 = |Z|$ by construction of Z. Say $Z = Z_1 + Z_2$. Let Z_1 be all triples (x, y, z) where x has degree 6, y has degree 6, u has degree 5 or 6 and w has degree 5. We can see that $q_5 = |Z_1|$. Then Z_2 must consist of all triples (x, y, z) where x has degree 5, y has degree 5, u has degree 5 or 6 and w has degree 5. But then $|Z_2| = q_6$. If we add up these we see that $q_5 + q_6 = |Z_1| + |Z_2| = |Z| = p_4$, which proves the statement.

Finally we will look at statement 4, which says $p_5 = q_7$. Again we will use a different kind of triple. This time (x, y, z) is called a triple if x, y and z are distinct neighbours of v, y is adjacent to x and z, u is the vertex not equal to v that is adjacent to x and y, w is the vertex not equal to x that is adjacent to u and y, and r is the vertex not equal to u that is adjacent to y and w, such that v, y and u have degree 6, w has degree 5 or 6, and r has degree 5. We call the sets of triples $|W|$. Again, by construction $p_5 = |W|$. If we look at the configuration in rule 5 we can see that there are 5 vertices adjacent to y. Because y has degree 6, there must be another vertex adjacent to y, which is z. But by definition of a triangulation, z must be adjacent to v and r. Now take a look at rule 7. We can see that q_7 is equal to all triples were the degree of z is 7 or higher. If z would not have degree 7 or higher it has degree 5 or 6. This would result in the following subgraphs:



Again, this cannot happen because both configurations are in the list (conf(1,3,5) and conf(1,3,7)) of good configurations. Therefore $q_7 = |W| = p_5$ and hence the statement is proven.

So we showed that all four equations hold and so the lemma is proven. □

Theorem 8. *Let G be an internally 6-connected triangulation. Let $v \in G$ be a vertex of degree 5 or 6 that after applying the discharging rules ends up with a charge $N > 0$, then a good configuration appears in the second neighbourhood of v.*

Proof. First, let v have degree 5. Assume no good configuration appears in the second neighbourhood of v, then the following equations hold:

1. $p_1 = q_2 + q_3$
2. $p_3 = q_4$
3. $p_4 = q_5 + q_6$
4. $p_5 = q_7$

Because v has degree 5 its initial charge is $10(6-5) = 10$. Also because v has degree 5, we know that the only rules that could affect v are rules 1,2 and 3, and for rule 2 and 3 v can only be the source. Therefore we can calculate the charge N as follows.

$N = 10 + p_1 - q_1 - q_2 - q_3$

Since $p_1 = q_2 + q_3$ we can rewrite this as:

$N = 10 - q_1$.

But since v has degree 5 by rule 1 it sends 2 charge to all vertices adjacent to v that have degree 5 or higher. Since all vertices in G have degree 5 or higher rule 1 sends 2 charge to all neighbours of v. Since v has degree 5 it has 5 neighbours and therefore:

$N = 10 - 5 \cdot 2 = 0$

This is a contradiction because $N > 0$. Therefore a good configuration appears in the second neighbourhood of v.

Let v have degree 6. Then the initial charge of v is $10(6-6) = 0$ and v can appear as sink in rules 1,3,4 and 5, and as source in rules 2,3,4,5,6 and 7. Therefore we can calculate N as follows:

$N = p_1 + p_3 + p_4 + p_5 - q_2 - q_3 - q_4 - q_5 - q_6 - q_7$

Now since $p_1 = q_2 + q_3$, $p_3 = q_4$ and $p_4 = q_5 + q_6$ and $p_5 = q_7$ we end up with:

$N = 0$

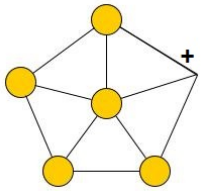
This is a contradiction because $N > 0$. Therefore a good configuration appears in the second neighbourhood of v. □

5.3 Case 2: v has degree 12 or higher

For this case we will first prove the following lemma:

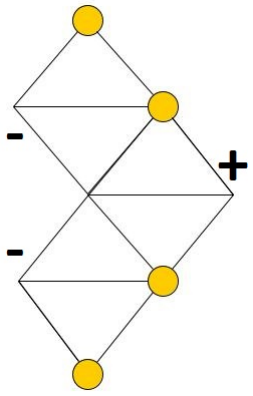
Lemma 8. *Let v be a vertex of degree 12 or higher of G that after applying the discharging rules ends up with positive charge. Let w be a fixed neighbor of v. Assume no good configuration appears in the second neighbourhood of v. Then the sum of $r(P)$ over all passes with source w and sink v is at most 5.*

Proof. We define R_k as the sum of $r(P)$ over all passes obeying rule k with source w and sink v. Here w is a fixed neighbour of v. Now $R = R_1 + ... + R_{32}$. We can see that $R_k \leq 2$ for all k. This is true because rule 1 has charge 2, and all other rules have charge 1. Some rules can appear twice with the same source and sink by mirroring the configuration. Therefore R_k might be 2 for some of the other rules. First let the degree of w be 9 or higher, then $R = 0$ because there are no rules where the source has degree 9 or higher. Now let the degree of w be 5. Then $R = R_1 + R_2 + R_3$ since these are the only rules where the source can have degree 5. Then $R_1 = 2$. Furthermore we have to prove that $R_2 + R_3 \leq 3$. Now if R_2 and R_3 are both equal to 2 then then it would look like this:



But then good configuration conf(1, 1, 1) appears, so not both R_2 and R_3 can be 2, so we have proven $R \leq 5$.

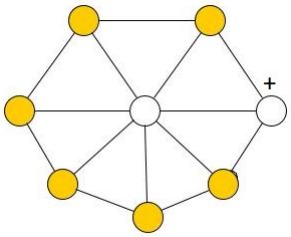
Now let the degree of w be 6. Then $R = R_2 + R_3 + R_4 + R_5 + R_6 + R_7$ because those are the only rules where the source can have degree 6. Now $R_3 + R_5 + R_6 \leq 2$ since if one of them appears at the top side of the edge between v and w, then none of the other two can appear. The same holds for the bottom side. We can conclude the same for rule 2 and 7, therefore $R_2 + R_7 \leq 2$. Furthermore $R_4 \leq 2$. Now $R > 5$ only if all three inequalities have equality. Thus $R_4 = 2$. But then $R_7 = 0$ because they cannot appear together. Therefore R_2 has to be 2. But if both R_2 and R_4 are equal to 2 then it would look like this:



There are four different possibilities for the vertices with the minus sign, but all those possibilities are in the list of good configurations (namely conf(1, 1, 3), conf(1, 1, 6) and conf(1, 2, 2)) and therefore they cannot appear. So if the degree of w is equal to 6 then $R \leq 5$.

Then, let the degree of w be 7. Let u_1 and u_2 be the two vertices adjacent to v and w. Let the degree of u_1 be c_1 and the degree of u_2 be c_2. Without loss of generality we can assume $c_1 \leq c_2$. We will look at multiple cases. Case 1:

Let $c_1 = c_2 = 5$, then $R = R_8 + R_9 + R_{10} + R_{11} + R_{12} + R_{13}$ because these are the only rules were where the degree of the source is 7 and both c_1 and c_2 can be equal to 5. Now $R_{10} \leq 1$. If $R_{10} = 1$ then $R_{11} = 0$, $R_8 \leq 2$, $R_9 \leq 2$, $R_{12} + R_{13} \leq 1$ since only one of them can appear at the same time. But now the only way that $R > 5$ is as equality holds for all inequalities. Now look at rule 9. If $R_9 = 2$ and $R_{10} = 1$ then it would look like this:



But this includes one of good configurations, conf(1,4,3), and can therefore not appear.

So $R_{10} = 0$. But then $R_{12} = 0$. Also $R_8 \leq 1$ because if it would appear twice rule 10 would apply too (remember that u_1 and u_2 have degree 5 by assumption). Furthermore, $R_9 \leq 2$, $R_{11} \leq 1$ and $R_{13} \leq 1$. So $R < 5$.

Case 2:

Let $c_1 = 5$ and $c_2 = 6$. Now $R = R_8 + R_9 + R_{14} + R_{15} + R_{16} + R_{17} + R_{18} + R_{19}$.

R_{16} and R_{18} are not both non-zero, and both are 0 if $R_8 = 2$. This means $R_8 + R_{16} + R_{18} \leq 2$. Also $R_9 \leq 1$ because again if rule 9 would appear twice there would appear some good configuration (conf(1, 4, 3) or conf(1, 4, 5)). Furthermore rule 14 and rule 19 cannot appear at the same time, so $R_{14} + R_{19} \leq 1$ and equivalently $R_{15} + R_{17} \leq 1$. So $R \leq 5$.

Case 3:

Let $c_1 = c_2 = 6$. Then $R = R_8 + R_9 + R_{20} + R_{21} + R_{22}$. Then $R_9 \leq 2$, $R_{20} + R_{22} \leq 1$ because they cannot appear at the same time, and $R_8 + R_{21} \leq 2$, because if rule 8 would appear twice, and rule 21 appears then a good configuration shows up, namely conf(1, 4, 3) or conf(2, 10, 6). Therefore $R \leq 5$.

Case 4:

Let $c_1 = 5$ and $c_2 = 7$. Then $R = R_8 + R_9 + R_{18} + R_{19} + R_{23} + R_{24} + R_{25}$. But with these constraints rule 8 and rule 9 can only appear once. So $R_8 \leq 1$ and $R_9 \leq 1$. Rule 18 and 19 cannot appear at the same time because of the degree of the sink, and therefore $R_{18} + R_{19} \leq 1$. Also only one of the rules 23, 24 and 25 can appear at the same time, so $R_{23} + R_{24} + R_{25} \leq 1$. Hence $R \leq 5$.

Case 5:

Let $c_1 = 6$ and $c_2 = 7$. Then $R = R_8 + R_9 + R_{21} + R_{22} + R_{26} + R_{27}$. This means rule 8 and rule 9 can only appear once, rule 21 and rule 22 cannot appear at the same time, and the same for rule 26 and 27. Therefore $R_8 \leq 1$, $R_9 \leq 1$, $R_{21} + R_{22} \leq 1$ and $R_{26} + R_{27} \leq 1$ and thus $R \leq 5$.

Since we handled all possibilities of degrees for c_1 and c_2 that appear in a rule we have proven that $R \leq 5$ if the degree of w is equal to 7.

Finally, we will look at the case that the degree of w is equal to 8. Then $R = R_{28} + R_{29} + R_{30} + R_{31} + R_{32}$. Only one of the rules 30, 31 and 32 can appear at the same time. Therefore $R_{30} + R_{31} + R_{32} \leq 1$. If one of them is non-zero then R_{28} and R_{29} are equal to 0 because the only combination that would be possible is rule 29 and rule 30, but then a good configuration would show up. Finally, since $R_k \leq 2$ for all k, we know that $R_{28} + R_{29} \leq 4$, so indeed $R \leq 5$.

Hence we have proven that for all possible degrees of w $R \leq 5$. □

Theorem 9. *Let G be an internally 6-connected triangulation. Let $v \in G$ be a vertex of degree 12 or higher that after applying the discharging rules ends up with a charge bigger than 0, then a good configuration appears in the second neighbourhood of v.*

Proof. Suppose no good configuration appears in the second neighbourhood of v. Let $d \geq 12$ be the degree of v. Then R, which is the sum of $r(P)$ over all passes with sink v, is at most 5. Let D be the set of all neighbours of v. Then $\sum_{w \in D} R(w) \leq 5d$. Now we calculate the charge N as follows:

$N = 10(6 - d) + \sum_{w \in D} R(w) \leq 60 - 10d + 5d = 60 - 5d \leq 0.$

This is a contradiction since we assumed $N > 0$, hence a good configuration appears in the second neighbourhood of v. □

5.4 Case 3: v has degree 7, 8, 9, 10 or 11.

Finally, we have to look at the cases where the degree of the vertex v that ends up with positive charge after applying the discharging rules is 7, 8, 9, 10 or 11. Unfortunately these cases have to be handled one by one, and furthermore the arguments are more difficult to understand than those of cases 1 and 2. In the 1997 proof a computer is used to prove this cases. The total length of these 5 proofs is about 13.000 lines, of which the authors say most require some thought to understand. In theory they are checkable by hand, but because of the length it is not doable. However, they wrote the program in such a way that a computer can check the proofs in a short amount of time (a couple of minutes). Also, two other people made programs to verify this case, independently from each other and the authors of the proof.

Now this concludes the proof of the Four Color Theorem. First we have shown that any minimal counterexample to the Four Color Theorem is an internally 6-connected triangulation. Then we made a set of 633 different good configurations. We showed that all these configurations are reducible, meaning they cannot appear in a counterexample to the Four Color Theorem. Finally, we showed that in every internally 6-connected triangulation at least one of the 633 good configurations appears. This means the minimal counterexample does not exist, and so no counterexample exist, meaning the theorem is true.

6 Conclusion

In this report, we have seen that the Four Color Theorem is a problem that dates back from 1852, when Frances Guthrie tried to color the map of England. The first notable attempt to prove the theorem was the proof of Kempe. Although there was an error in the proof, it did prove that a minimal counterexample cannot contain a vertex of degree 4 or less. Furthermore, with just a couple of small additions the Five Color Theorem could be proven. This was eventually done by Percy Heawood, who was also the man who in 1890 discovered the error in the proof of Kempe by giving a counterexample.

The proof of Kempe also introduced Kempe-Chains. These were used by Birkhoff in 1913 to prove that every minimal counterexample to the Four Color Theorem is an internally 6-connected triangulation. This result was used eventually by Appel and Haken, who with the help of Heesch and some others managed to give a correct proof in 1976. Later, in 1997, Robertson, Sanders, Seymour and Thomas gave an easier proof that was based on the same methods as the 1976 proof. These methods were reducibility, discharging and unavoidability. In both proofs a set of configurations was used. They proved that all configurations in the set were reducible, meaning they cannot appear in a minimal counterexample. Then with the help of the discharging method they managed to show that the set of configuration was unavoidable, meaning that every internally 6-connected triangulation contains at least one of the configurations.

Unfortunately, one has to use a computer to prove both reducibility and unavoidability, simply because it is so much work that it cannot be done by hand. All known proofs make use of reducibility, unavoidability and discharging and all of them use the computer for the proof. We can see reductions in the amount of work that the computer has to do. In 1976, a set of 1476 configurations was used. Furthermore, the discharging procedure used in this proof needed 486 rules. In 1997, these numbers were significantly lower. It needed "only" 633 configurations and 32 discharging rules. In 2010, John Steinberger[1] gave a proof in which 2822 configurations were used with 42 discharging rules, but all configurations were D-reducible. Although D-reducibility is much easier to show than C-reducibility this proof still used more computer time than the 1997 proof. The reason for this was not only the size of the set, but also the size of the required configurations.

So far nobody has been able to prove the theorem without the use of the computer, and even with the computer it is very difficult to understand. This is in a way strange given how close Kempe got with only very easy arguments. If one would look further into the proof of the theorem one could either try to find a proof that does not make use of the computer or try to find a way to reduce the work the computer needs to do. Both will not be easy, and the question is whether it is even possible. Also the question is whether it is worth it. I don't think it is worth the time to try to improve the computer proof. However, if someone could find a proof that does not need a computer, it would be a great result.

References

[1] Steinberger, J. (2010). An unavoidable set of D-reducible configurations. Transactions of the American Mathematical Society, 362(12), 6633-6661.

[2] Thomas, R. (1998). An update on the four-color theorem. Notices of the AMS, 45(7), 848-859.

[3] Davidson, R. The 150 Year Journey of the Four Color Theorem.

[4] Soifer, A. (2009). Kempe–Heawood's Five-Color Theorem and Tait's Equivalence. In The Mathematical Coloring Book (pp. 176-186). Springer, New York, NY.

[5] Robertson, N., Sanders, D., Seymour, P., Thomas, R. (1996). A new proof of the four-colour theorem. Electronic research announcements of the American Mathematical Society, 2(1), 17-25.

[6] Robertson, N., Sanders, D., Seymour, P., Thomas, R. (1997). The four-colour theorem. journal of combinatorial theory, Series B, 70(1), 2-44.

[7] Kempe, A. B. (1879). On the geographical problem of the four colours. American journal of mathematics, 2(3), 193-200.

[8] Birkhoff, G. D. (1913). The reducibility of maps. American Journal of Mathematics, 35(2), 115-128.

[9] Heawood, P. J. (1898). On the four-colour map theorem. Quart. J. Pure Appl. Math., 29, 270-285.

[10] Appel, K., Haken, W. (1976). Every planar map is four colorable. Bulletin of the American mathematical Society, 82(5), 711-712.